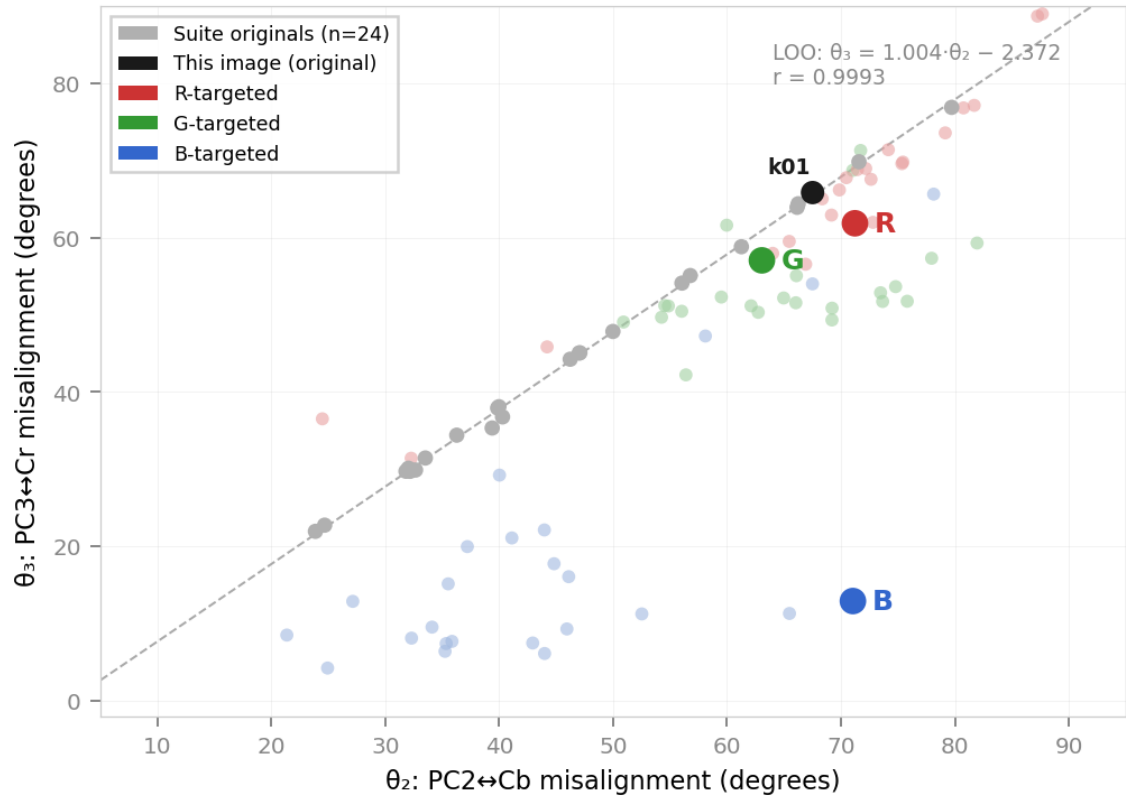


Compression Profile: KODIM01 768x512 | 24-bit RGB | PNG (lossless)

stone building | Strongly One-Dimensional (PC1 = 94.36%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.8627	0.8400	0.9914	0.898	67.51°	65.87°	+0.46°
R-targeted	0.0133	-0.0111	0.9547	0.326	71.23°	61.89°	-7.25°
G-targeted	-0.1991	0.9667	-0.1552	0.440	63.06°	57.08°	-3.86°
B-targeted	0.9479	-0.0419	-0.0735	0.354	71.05°	12.9°	-56.06°

R-targeted isolates red (R-G, R-B < 0.01) while preserving G-B coupling at 0.955.
G-targeted isolates green (R-G, G-B < 0.20) while preserving R-B at 0.967.
B-targeted preserves R-G at 0.948 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold (r = 0.9993).

Compression Profile: KODIM01768x512 | 24-bit RGB | PNG (lossless)

stone building | Strongly One-Dimensional (PC1 = 94.36%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

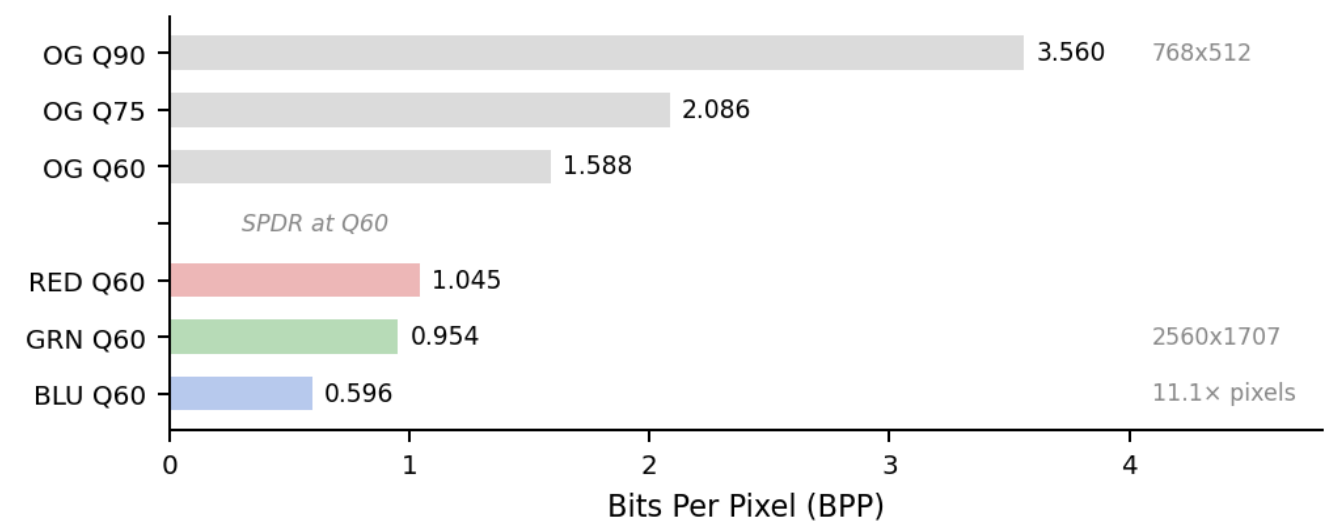
ORIGINAL (kodim01)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	719.2 KB	14.984	393,216
JPEG Q90	JPEG	768x512	170.9 KB	3.560	393,216
JPEG Q75	JPEG	768x512	100.1 KB	2.086	393,216
JPEG Q60	JPEG	768x512	76.2 KB	1.588	393,216

REDISTRIBUTED (kodim01 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	557.2 KB	1.045	4,369,920
GRN Q60	JPEG	2560x1707	509.1 KB	0.954	4,369,920
BLU Q60	JPEG	2560x1707	318.0 KB	0.596	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1x the original pixel count.

Compression Profile: KODIM01 768x512 | 24-bit RGB | PNG (lossless)

stone building | Strongly One-Dimensional (PC1 = 94.36%)

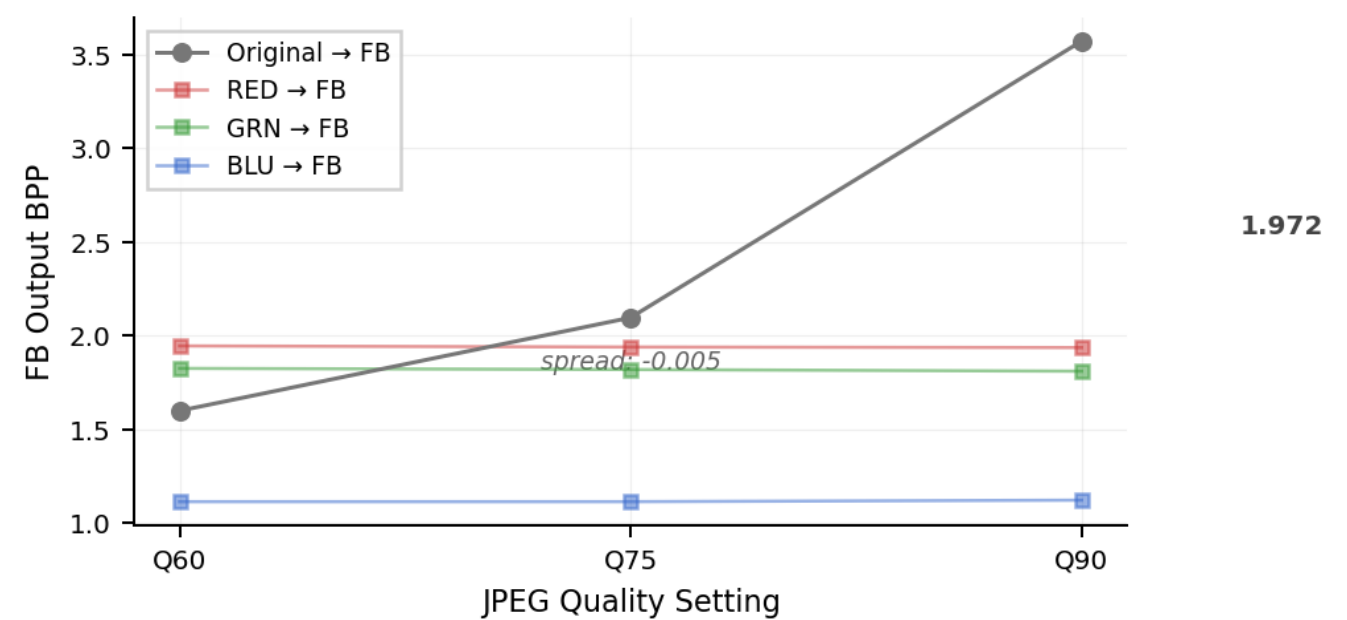
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	3.474	—
R-targeted → FB	2560x1707	2048x1366	1.935	44.3%
G-targeted → FB	2560x1707	2048x1366	1.809	47.9%
B-targeted → FB	2560x1707	2048x1366	1.121	67.7%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.414	59.3%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.8943 vs 0.898 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.972 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.005 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim01

Best BPP (Q60, direct)	0.596	2560x1707, 4,369,920 px
Best BPP through FB	1.121	2048x1366 output
vs OG Q90 → FB	68.6%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216